

STETTLER AGDM, AB, Canada

WMO: 712890

Lat: 52.3469N

Lon: 112.5956W

Elev: 2608

StdP: 13.36

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.7	-14.8	-26.1	1.4	-19.3	-20.9	1.9	-14.7	27.3	21.0	24.3	20.6	8.2	190	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	84.4	62.8	80.9	61.1	77.4	60.0	65.9	78.1	63.7	76.2	62.0	73.8	9.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	89.7	71.6	59.2	82.7	68.5	57.2	76.9	66.6	32.1	77.5	30.4	76.5	29.0	73.9	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	21.7	19.0	DB	-26.0	90.2	5.3	3.7	-29.8	92.9	-32.9	95.0	-35.9	97.1	-39.7	99.8	(4)
(5)			WB	-25.6	70.4	5.8	2.6	-29.8	72.3	-33.1	73.8	-36.4	75.3	-40.6	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	16.1	14.3	25.4	39.0	50.9	57.8	62.7	60.5	52.1	40.1	25.0	14.5		
	DBStd	20.62	14.19	13.76	13.38	9.53	7.80	5.57	5.30	6.12	8.20	9.17	12.51	13.75		
	HDD50	5494	1051	1000	763	342	85	6	0	3	71	324	750	1099		
	HDD65	9804	1516	1420	1227	779	440	224	108	164	393	770	1199	1564		
	CDD50	1240	0	0	1	13	111	241	393	329	133	18	1	0		
	CDD65	77	0	0	0	0	2	9	37	25	4	0	0	0		
	CDH74	1558	0	0	1	5	97	171	619	478	180	7	0	0		
	CDH80	385	0	0	0	0	12	26	175	124	48	0	0	0		
	WSAvg	9.4	10.0	9.5	9.7	10.4	10.4	9.5	8.1	7.8	8.7	9.8	9.5	9.7		
	PrecAvg	16.7	0.8	0.5	0.7	1.2	1.9	3.1	3.0	2.2	1.3	0.7	0.6	0.6		
(15) Precipitation	PrecMax	20.8	2.5	1.0	1.5	2.4	3.3	4.5	6.5	4.2	3.0	1.8	2.2	1.2		
	PrecMin	11.4	0.0	0.1	0.1	0.2	0.6	1.0	0.9	0.2	0.1	0.2	0.1	0.2		
	PrecStd	2.8	0.6	0.3	0.4	0.6	0.8	1.0	1.3	0.9	0.8	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	43.6	57.9	72.0	81.7	83.1	88.9	88.0	85.9	74.0	56.9	44.9		
		MCWB	36.7	36.8	44.2	50.6	56.2	61.3	64.9	62.9	59.8	53.5	44.7	38.0		
	2%	DB	40.6	39.1	50.6	66.1	76.5	78.8	84.8	83.4	79.1	65.3	49.2	40.4		
		MCWB	35.4	33.9	41.0	48.6	54.1	60.1	65.1	62.0	57.7	50.1	40.8	35.1		
	5%	DB	38.0	35.9	45.0	60.3	72.2	74.9	80.9	79.4	73.8	59.5	43.5	36.6		
		MCWB	33.7	32.0	37.7	44.9	52.4	58.2	63.2	61.6	55.8	46.9	36.8	32.2		
	10%	DB	34.6	32.8	40.7	55.2	67.5	71.2	76.8	75.4	68.0	54.3	39.9	32.8		
		MCWB	30.9	29.7	35.4	42.9	50.4	56.6	62.1	60.3	53.4	44.0	34.5	29.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	37.2	45.9	52.5	57.8	64.9	70.0	67.7	61.4	54.5	45.8	38.4		
		MCDB	42.5	42.6	55.7	69.6	76.0	77.4	83.0	79.6	79.3	70.9	56.2	44.0		
	2%	WB	35.7	34.2	41.2	49.2	55.5	62.1	67.0	64.8	58.8	50.9	41.1	35.3		
		MCDB	40.2	38.3	49.4	64.0	72.6	74.5	79.5	77.6	76.0	63.5	48.0	39.9		
	5%	WB	33.8	32.2	38.0	46.2	53.6	60.0	64.7	62.8	56.6	47.7	37.6	32.3		
		MCDB	37.9	35.6	44.4	58.5	69.4	71.2	77.2	76.0	71.1	57.9	42.7	36.3		
	10%	WB	31.1	29.8	35.6	43.5	51.6	58.0	63.0	61.0	54.4	44.9	35.0	29.3		
		MCDB	34.6	32.7	40.5	53.8	65.1	68.6	74.5	73.0	67.2	53.4	38.9	32.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.2	17.5	20.8	23.3	21.0	23.4	23.7	23.2	20.5	16.2	15.6		
		MCDBR	17.5	18.9	21.9	30.9	31.0	27.5	30.9	31.6	33.5	28.7	20.8	17.4		
		MCWBR	13.8	15.1	13.9	15.7	13.3	12.0	13.9	14.2	15.0	15.5	13.6	13.6		
		MCDBR	17.4	18.5	20.7	28.7	27.7	24.1	27.1	27.8	30.6	27.8	19.1	17.2		
	5% WB	MCWBR	13.9	15.0	13.8	15.5	12.3	12.0	13.8	13.9	14.4	15.3	13.2	13.8		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.272	0.290	0.328	0.348	0.358	0.354	0.367	0.314	0.284	0.256	0.246		
	taud		2.284	2.374	2.428	2.366	2.360	2.366	2.407	2.349	2.469	2.531	2.464	2.322		
	Ebn at Noon		233	272	290	288	286	284	283	272	277	267	244	224		
	Edh at Noon		20	24	28	34	36	36	35	35	27	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		344	676	1120	1499	1824	1931	2027	1654	1150	687	352	253		
	RadStd		31	56	84	138	115	122	105	114	124	81	35	25		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (86 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page